

Peek – Shape – Grab: A Methodology in Three Stages for Approximating the Non-dominated Points of Multiobjective Discrete/Combinatorial Optimization Problems with a Multiobjective Metaheuristic

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Abstract. The paper discusses on the role of components of multiobjective metaheuristics in the context of multiobjective discrete/combinatorial optimization. It suggests to separate in three stages the design of an operational solver for this class of optimization problems, featuring a methodology in this context. The arguments are illustrated using the knapsack problem as support, along numerical experiments.

Keywords: Multiobjective optimization · Multiobjective metaheuristic · Discrete and combinatorial problems · Algorithmic design methodology

1 Introduction

1.1 Context and Motivations

To solve exactly a Multiobjective Optimization Problem (MOP), without introducing (a priori or interactively) the preferences of a decision maker, means to compute the set of non-dominated points denoted by Y_N . In front of a difficult MOP to solve, a reasonable alternative to exact methods is to derive an approximation method. A Multiple Objective MetaHeuristic (MOMH) is an approximation method in MOP context, which provides a good tradeoff between a high quality approximation of Y_N , and the time and memory requirements to obtain it.

Evolutionary MultiObjective (EMO) algorithms have been originally introduced for approximating Y_N of multiobjective non linear optimization problems with continuous variables (MONLP). A quick look on the pioneer papers ([12, 30, 31] by example among many others papers), on the standard benchmarks (e.g. ZDT, DTLZ, WFG, all of them can be found online here [20]), and on the

examples developed into reference books (e.g. [1,2]) confirms this claim. And when the multiobjective discrete/combinatorial optimization (MODCO) side is considered, the knapsack problem is mainly used as benchmark ([19,33], etc.).

It is then not surprising to find specific mechanisms within EMO algorithms for dealing with particularities of MONLP. The filtering of approximated solutions is one illustration of such a mechanism. As the variables are continuous, and EMO algorithms handle discrete values on these variables, the filtering aims to prevent a huge number of discrete values, corresponding to solutions, and moreover, which may over represent a sub set of the non-dominated points searched.

However, such a filtering mechanism is in general not relevant for MODCO where the aim here is to grab the set of non-dominated points or its approximation. A similar observation can be done about the initial population of solutions provided in input of a EMO algorithm. For several reasons, this population is generally composed of randomly generated solutions. However, useful information on the optimization problem to solve, as bounds or bounds sets, may be generally obtained and exploited for MODCO [14].

For situations where the optimization problem to deal with belong to the class of MODCO, EMO algorithms have been coupled with several usual well known techniques in the field of discrete/combinatorial optimization, such as, chronologically Local Search (LS) [24], Pareto Local Search (PLS) [23], 2 Level Local Search (2LLS) [17], etc. The literature presents these blended algorithms under the terminology of “memetic algorithms” (heuristic and local search), “hybrid algorithms” (heuristic mixed with an other exact or heuristic technique), and more recently by “matheuristics” (when exact or heuristics are collaborating within a same solving process).

Different studies have proposed several coupling (such as local search, or again path relinking) and specific mechanisms of local search have been proposed ([4,14,21], etc.). But the introduction of these components has progressively modified the role of the EMO algorithm, itself now viewed as a component of the MOMH.

This paper discusses about the role of each component in the context of MODCO. On basis of contributions that we did to the field of multiobjective metaheuristics since [16], we suggest to separate in three stages the design of a MOMH for MODCO, featuring a methodology in this context. To follow the tradition, the arguments are illustrated using the knapsack problem as support, along numerical experiments.

The rest of the paper is organized as follow. After a brief introduction of notations and definitions in the next subsection, Sect. 2 presents the methodology, first in sketching the main conclusions who led to that proposal, and second, the three stages of the methodology. Section 3 illustrates the methodology with support of numerical examples and mentions two real-life case studies managed with respect to the methodology.

1.2 General Definitions and Notations

Let $\mathbb{R}^n$ and $\mathbb{R}^p$ be Euclidean vector spaces referred to as the decision space and the objective space. $x \in \mathbb{R}^n$ is a decision. Let $X \subset \mathbb{R}^n$ is a feasible set and let f be a vector-valuated objective function $f : \mathbb{R}^n \rightarrow \mathbb{R}^p$ composed of p real-valued objective functions, $f = (f_1, \dots, f_p)$, where $f_k : \mathbb{R}^n \rightarrow \mathbb{R}$ for $k = 1, \dots, p$. A multiobjective optimization problem is defined as

$$\min \{(f_1(x), \dots, f_p(x)) : x \in X\}. \quad (\text{MOP})$$

This paper addresses MultiObjective Discrete/Combinatorial Optimization problems, which can be formulated as

$$\min \{Cx : Ax \geq b, x \in \mathbb{Z}^n\}. \quad (\text{MODCO})$$

Here C is a $p \times n$ objective function matrix, where c^k denotes the k -th row of C . A is an $m \times n$ matrix of constraint coefficients and $b \in \mathbb{R}^m$. Usually the entries of C , A and b are integers. When the feasible set $X = \{Ax \geq b, x \in \mathbb{Z}^n\}$ describes a combinatorial structure, we talk about MultiObjective Combinatorial Optimization problems (see [6]), otherwise we talk about MultiObjective Discrete Optimization problems.

By $Y = f(X) := \{f(x) : x \in X\} \subset \mathbb{R}^p$ we denote the image of the feasible set in the objective space. We consider optimal solutions of (MODCO) in the sense of efficiency, i.e., a feasible solution $x \in X$ is called efficient if there does not exist $x' \in X$ such that $f_k(x') \leq f_k(x)$ for all $k = 1, \dots, p$ and $f_k(x') < f_k(x)$ for some k . In other words, no solution is at least as good as x for all objectives, and strictly better for at least one.

Efficiency refers to solutions x in decision space. In terms of the objective space, with objective vectors $f(x) \in \mathbb{R}^p$ we use the notion of non-dominance. If x is efficient then $f(x) = (f_1(x), \dots, f_p(x))$ is called non-dominated. The complete set of efficient solutions is X_E , the set of non-dominated points is Y_N . We may also refer to Y_N as the non-dominated frontier. Also, a set of potentially efficient solutions is denoted by X_{PE} and Y_{PN} , respectively in the decision and objective space. More details about definitions and notations are available in [5, 8].

2 Methodology in Three Stages

The proposed methodology arises from the major conclusions raised in our previous research since the end of the nineties. The main lines of these conclusions are sketched in this subsection.

2.1 Observations Leading to the Methodology Definition

- **Bound sets on Y_N** [7]. The concept of bound sets has been mentioned in the eighties by Villarreal et al. [32], but it was only developed for the first time in 2001 [7] and fully defined in 2007 [9]. The lower and upper bound sets are useful as criterion for deciding when a solution is promising and then to

trigger -e.g.- a local search from that solution (see [17]). An heuristic version of lower bound set has been introduced in [26], as a way to enlarge the area corresponding to promising solutions.

- **Local search and seeding solutions** [22]. The resolution of the biobjective scheduling problem defined by $1 \parallel (\sum C_i, T_{max})$ has been considered with a specific multiobjective genetic algorithm (named MGK) coupled with a local search acting in two steps. One main contribution of the paper concerns the “seeding solutions”, as principle for accelerating the convergence of the algorithm. The cases (i) without seeding solutions, (ii) with exact seeding solutions and (iii) with an approximation of the seeding solutions are analyzed. A component based on a two steps local search is also integrated. As conclusion of this paper, the positive impact of the seeding solutions is clearly stated.
- **Path-relinking and genetic information** [14]. For the linear assignment problem with two objectives, a path-relinking operator has been embedded for the first time in a population based-method. Seeding solutions, in particular represented by the set of supported solutions which can be easily computed for this combinatorial problem, are intensively used by the method. The conclusion here raises that useful information may be derived from the seeding solutions, and may advantageously be used by the method - notably by the path relinking- for speeding up drastically the convergence of the algorithm.
- **Greedy randomized initial solutions** [3]. Greedy Randomized Adaptive Search Procedure (GRASP) [10,11] is a well-known metaheuristic who has shown its efficiency on several hard combinatorial optimization problems. One proposal developed in the cited paper consists in the use of the first phase of GRASP for computing a subset of initial solutions, presented afterwards in input of a EMO algorithm (a modified “Strength Pareto Evolutionary Algorithm” [34] has been considered in the paper). The proposal has been experimented on set packing problems. In conclusion, in mixing quality and diversity within solutions built, GRASP permits to elaborate very quickly a large population of very good individuals with a control on their distribution into the objective space.

Several of these proposals have been considered in recent papers; some of them are cited here. Haubelt et al. [18] initialize the population in order to guide the search towards the feasible region. They proposed the incorporation of a method called Pareto-Front-Arithmetics for constructing initial populations into existing EMO algorithms. Pasia et al. [26] use a single objective “Ant Colony Optimization (ACO)” algorithm for generating individuals objective by objective for a flowshop problem. [25] proposes a two-phase MOMH. Phase one consists of a weighted sum-based solution method belonging to the “Variable Neighborhood Search (VNS)” metaheuristic. Phase two of is a heuristic module based on Path-relinking. [21] proposes a Two-phase Pareto local search, combining seeding solutions and PLS, for the biobjective traveling salesman problem.

2.2 The Methodology

The proposed methodology is structured in three stages entitled “Peek–Shape–Grab (PSG)”. Each stage is characterized by a well stated mission, and is concretized by an algorithmic principle. Of course, each algorithmic principle may be instantiated in different ways. Section 3 presents how the three stages have been instantiated in the experiment reported. It is obvious that an instantiation may be replaced by another one, as long as it complies with the mission of the methodology stage.

1. Peek

The aim of this stage is to probe Y_N in order *to construct a good initial population* (Fig. 1, top). The quality expected by this population is to represent *a good coverage* of the non-dominated frontier Y_N , even if the approximation is sparse due to the local character of the operator applied. It is expected from the resulting population to act as attractor for the approximation mechanism during the next stages. Very high quality solutions are then not specially expected, preferring good but diversified solutions.

Input: A numerical instance for the considered MODCO problem.

Output: A non-connected approximation of Y_N which represents of a good initial population, i.e. a representative coverage of Y_N .

2. Shape

The aims of this second stage is to spread the population along Y_N in order *to get a well distributed population* without missing subparts of the non-dominated frontier (Fig. 1, middle). The quality expected is now to deliver a population connected and widespread along Y_N . The generation of a global approximation exploiting genetic information available is typically the strength of genetic algorithms, which have the advantage to explore the search space.

Input: A constructed population returned at the end of the stage “Peek”.

Output: A population giving a global approximation of Y_N and uniformly distributed.

3. Grab

The last stage aims to improve in an aggressive way the approximation, using all information collected during the search process until now, in order *to deliver a high quality approximation of Y_N* (Fig. 1, bottom). Individuals of the population are locally improved in order to shift the population to “its (local) optimum”. Local improvement is a matter of local search procedure bracketed by upper/lower bound sets and guided by components like path-relinking.

Input: A population uniformly distributed giving the shape of Y_N

Output: A high quality approximation of Y_N .

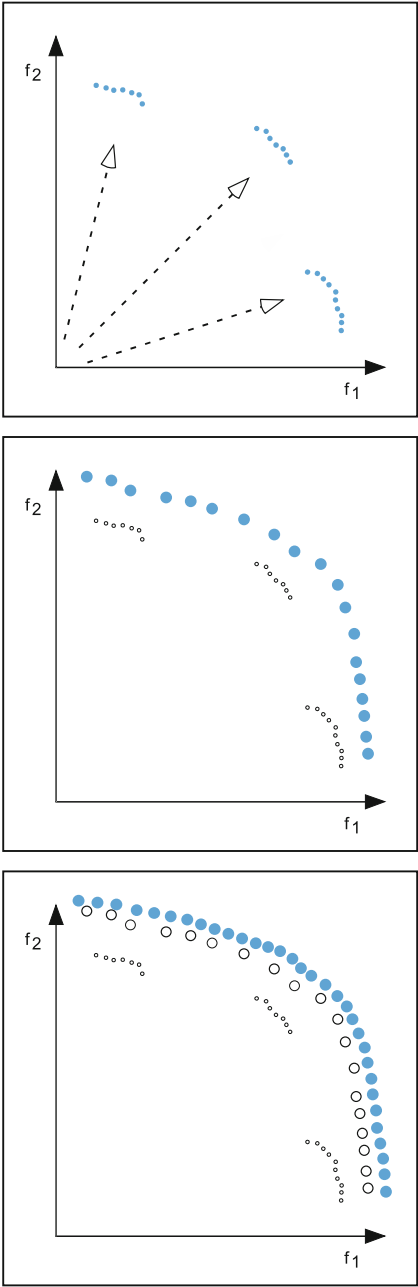


Fig. 1. The figure illustrates the awaited effect (depicted by plain bullets) of *peek* (top), *shape* (middle), and *grab* (bottom) stages. The empty bullets recall the populations presented in input of the preceding stages.

3 Illustrations of the Methodology

3.1 The Version of Knapsack Problem Serving as Support for the Illustration

The MODCO problem is the biobjective ($p = 2$) uni-dimensional ($m = 1$) binary knapsack problem:

$$\left[\begin{array}{ll} \max \sum_{j=1}^n c_j^k x_j & k = 1, \dots, p \\ s.t. \sum_{j=1}^n w_{ij} x_j \leq \omega_i & i = 1, \dots, m \\ x_j \in \{0, 1\} & j = 1, \dots, n \end{array} \right]$$

The computer used for conducting our experiment has these characteristics: intel Core 2 Duo E8500 @ 3.16 GHz; 3.4 GB RAM. The operating system is Ubuntu, and the algorithm is implemented in C++. The instance used is the 1B/2KP500-1A (serie 1B; 2 objectives; 500 variables; random coefficients) available on the MOCOLib¹.

3.2 About the Impact of Seeding Solutions [15]

Figure 2 shows experimentally the impact of seeding solutions on the convergence of the Y_{PN} approximation. In that experiment, the definition of the solution algorithm is fixed and consists in a basic multiobjective genetic algorithm. Its components and parameters (crossover and mutation operators, the probability of crossover and mutation, the size of the population, number of generations) are also freezed. Starting from an initial population composed by the same number of individuals, but built in three different manners, the experiment shows the evolution, in comparison to Y_N (black bullets). That evolution is examined before the first generation (red bullets), after 10 generations (green bullets), and after 50 generations (blue bullets).

- In the first case (top of Fig. 2), the population is only composed of individual randomly generated.

MOGA algorithm reacts as expected, the population of individuals evolves well towards the non-dominated frontier. However, we remark that the final population returns an approximation that describes only in the central part of non-dominated frontier, and this without reaching Y_N . Therefore, the approximation does not provide a valuable indication of all the non-dominated frontier.

- In the second case (middle of Fig. 2), three seeding solutions are built. They correspond to three solutions computed by application of the following simple greedy constructive algorithm. The convex combination of the two objectives

¹ <http://xgandibleux.free.fr/MOCOLib/>.

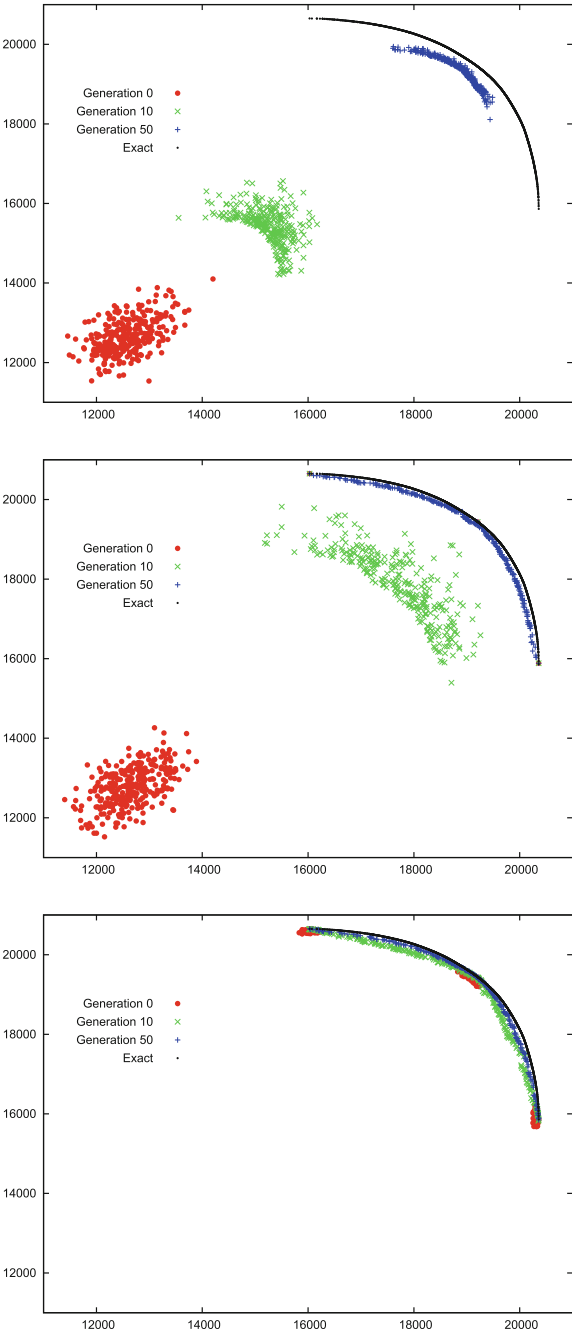


Fig. 2. The figure illustrates different states of the populations along the generations, for a same computing budget, a same tuning of parameters, and three variants of seeding solutions: (top) none seeding solution, (middle) with 3 seeding solutions plus random solutions, (bottom) with 3 sets of greedy solutions. The axes x and y represent respectively f_1 and f_2 . (Color figure online)

$f_\lambda = \lambda f_1 + (1 - \lambda)f_2$ is set, giving a parametrized function by λ . The three solutions are obtained by optimizing the resulting single objective problem for the values $\lambda = \{0; 0.5; 1\}$ with a greedy algorithm. Thus, two solutions correspond to the sub-optimal solutions when objectives are optimized separately ($\lambda = 0$ and $\lambda = 1$), the third corresponds to a compromise solution ($\lambda = 0.5$). The rest of the population is only composed by individuals randomly generated.

For the same computing budget given to the algorithm, we observe by comparison with the previous case that the population is closer to Y_N and better spread already after 10 generations. The difference is especially striking at the end of the 50 generations, where the final population gives a good description of the entire non-dominated frontier. Y_{PN} is indicative of Y_N but remains improvable. Nevertheless, the approximation Y_{PN} is able to find several points of Y_N in the neighborhood of the seeding solutions.

- In the third case (bottom of Fig. 2), three sets of seeding solutions are built and constitute exclusively the initial population; here, there is therefore no random solution. These solutions are obtained by application of the stage “peek” presented in Sect. 2, based on the constructive phase of GRASP. The optimized function is the parametric function by defined as in the previous case, with the values $\lambda = 0$, $\lambda = 0.5$, and $\lambda = 1$. All the solutions generated are gathered for constituting the initial population.

For the same computing budget given to the algorithm, the three sets of individuals, which are initially not connected together, are now uniformly distributed along the non-dominated frontier providing only one set of individuals, representative of a good global approximation, already at the tenth generation. At the end of the fiftieth generation, the description of Y_N by Y_{PN} is improved and more points of Y_N are reached, in comparison with the previous case. Y_{PN} is representative of Y_N and “matches” with Y_N .

In preliminary conclusion, we observe that the seeding solutions (1) speed up the convergence of the population of individual toward the non-dominated frontier and (2) incite the individuals in the population to be well distributed along the non-dominated frontier. The saving on the computation budget achieved during the convergence of the population thanks to the seeding solutions allows to relocate the saved time on the others phases of the algorithm. The evolutionary multiobjective component is awaited as component for spreading the individuals of the population, and not any longer as component for generating a high quality approximation. By the way, the construction cost of the population with the proposed principle is insignificant in comparison to the required computing time by the two others phases. That experimental observation validates the suitability of the first phase of the methodology, and the mission assigned to the second phase.

3.3 Experimental Demonstration [15]

1. Peek with GRASP

Principle: The 2 objective functions $f_k, k = 1, 2$ for the considered MODCO problem are aggregated using a weight vector λ into a parametric function $f_\lambda = \phi(f_1, f_2, \lambda)$. This function f_λ is then optimized with the construction step of GRASP for a collection of values λ . The utility function defined for ranking the items of the knapsack is the ratio profit/weight.

2. Shape with MOGA

Principle: The biobjective knapsack problem is tackled with a EMO solver. Here, a simple homemade MOGA has been designed. Obviously, all recent advances on EMO based algorithms and all outcomes of recent studies related on solving the knapsack problem by a EMO algorithm are welcome for strengthening this stage.

3. Grab with IPLS

Principle: A selection of solutions are optimized on the objectives f_1, f_2 with a method belonging to the iterated Pareto local search (IPLS). In a short, a niche of size σ allows to select candidates for a local search. A lower bound set is used for deciding if the improvement is triggered. A recursive principle is performed while new potential non-dominated points are grabbed.

The algorithm has been setup with a computing budget of $T = 120$ s dispatched as follow:

$$\text{GRASP}(\approx 0) + \text{MOGA}\left(\frac{T}{2}\right) + \text{IPLS}\left(\frac{T}{2}\right)$$

The resolution has been repeated 10 times. Figure 3 shows the effect of the three stages, in plotting the populations of individuals in the objective space. Again we can notice that the points issued from the first phase (red bullets) are well spread after the second phase (green bullet). The outputs of the second and third phases are visually indistinguishable.

The quality indicator $M1$ is thus computed. It measures the percentage of exact non-dominated points of Y_N detected in Y_{PN} , i.e.

$$M1 = \frac{|Y_{PN} \cap Y_N|}{|Y_N|}$$

The values are given in average upon the 10 runs, and also the minimal and maximal values collected. The value of $M1$ moves from 30.75% after the phase 2, to 77.76% after the phase 3. The positive effect of the last phase of the methodology do not leave any doubt. Moreover, according the minimal and maximal values collected, the algorithm appears robust as the interval is tight.

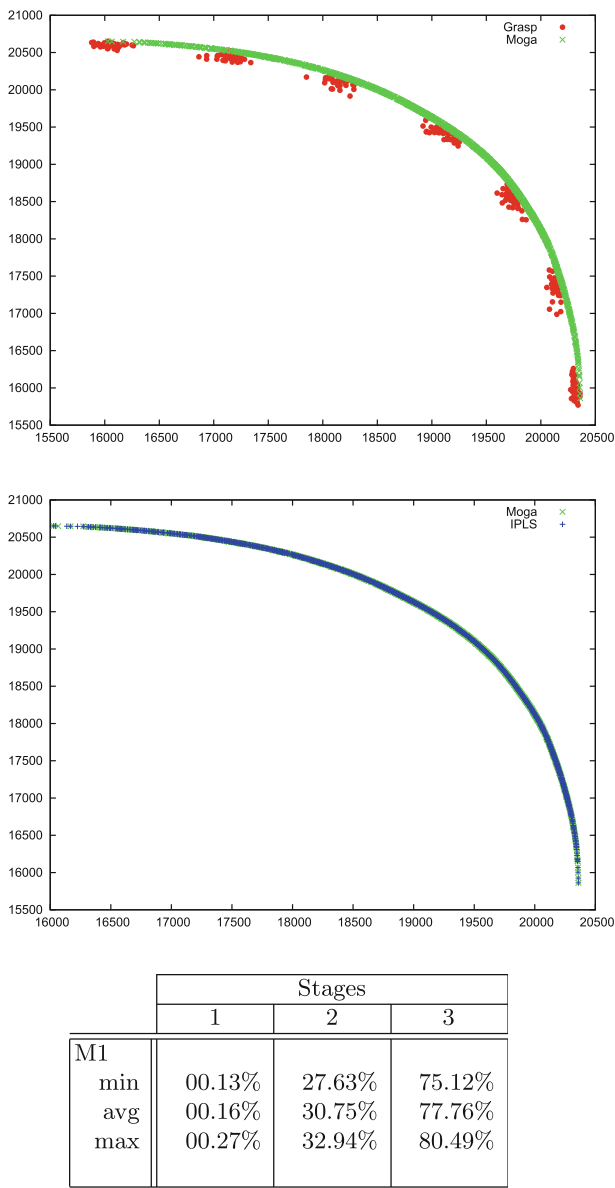


Fig. 3. Dump of Y_{PN} at the end of the first and second stage (top) – Dump of Y_{PN} at the end of the second and third stage (bottom). The axes x and y represent respectively f_1 and f_2 . The table report the measure of M1 (percentage min/avg/max of exact non-dominated points detected) at the end of the three stages.

3.4 Two Real Life Problems

- **A biobjective telecommunication network expansion planning problem** [13]. This case study comes from a research collaboration with FranceTelecom. The problem concerns a network expansion planning problem in telecommunication. The mathematical model formalizing this problem belongs to the class of biobjective facility location problems with mixed integer variables. The seeding solutions are obtained with a branch-and-bound algorithm limited in computing time, and embedded into a ϵ -constraint method. The EMO solver is NSGA-II modified (without filtering procedure). A local search and a path-relinking finalize the solver.
- **A multiobjective machine reassignment algorithm for data centers** [27–29]. This work is considered seminal in the data center management community and has led to a collaboration with IBM [27, 29]. The problem consists in reassigning virtual machines in large data centers – hence a large and constrained multiobjective problem with mixed variables. We use a GRASP algorithm to get an initial population, followed by NSGA-II and PLS.

4 Conclusion and Open Questions

After having studied and better understood the behavior of MOMH presented since the mid-nineties, identified the strengths and weaknesses, this paper notifies on the importance of a non-monolithic approach to design a resolution method for a MODCO. It therefore presents a methodology of design in three phases, Peek-Shape-Grab. It illustrates its suitability using a very simple algorithmic instantiation but yet effective, when computational resources are limited, and according to the collected observations and measures. In accordance with the definition of a Metaheuristic, this methodology suggests missions and proposes algorithmic directions. For each MODCO problem, the three phases have to be instantiated and to integrate the known results from the literature in order to benefit of last advances and give rise has an operational effective resolution algorithm.

Several questions remain to be addressed, for example on the implementation of the phase 3: local search on all available points or not? How to select a relevant subset of points? How to move forward in the local search without losing information? Also, what questions and answers to bring when the problem rises in the number of objectives to deal with simultaneously? Up to how many objectives it is reasonable to suggest this methodology? So many questions to investigate into future researches.

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